

1. Administrative & Study Identification

Full Study Title: A Phase 2/3 Randomized, Placebo-Controlled, Double-blind, Clinical Study to Evaluate the Efficacy, Safety, and Pharmacokinetics of Vericiguat in Pediatric Participants with Heart Failure due to Systemic Left Ventricular Systolic Dysfunction (VALOR) **Registry Title:** Efficacy, Safety, and Pharmacokinetics of Vericiguat in Pediatric Participants With Heart Failure Due to Left Ventricular Systolic Dysfunction (MK-1242-036) **Short Title/Acronym:** VALOR (MK-1242-036) **Protocol Number:** MK-1242-036 **NCT Number:** NCT05714085 **Posting ID:** 791002599937791567 **Trial Status:** Recruiting **Sponsor/Company:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Vericiguat (Verquvo) / MK-1242 (Tablet and Oral Suspension) | **ATC Code:** C01DX22 **Phase of Development:** Phase 2/Phase 3 **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To compare the efficacy of vericiguat versus placebo on the change in N-terminal pro-brain natriuretic peptide (NT-proBNP) from baseline to Week 16 of the Base Period. The primary hypothesis is that vericiguat is superior to placebo in reducing NT-proBNP at Week 16. **Secondary Objectives:** To evaluate the safety, tolerability, and pharmacokinetics of vericiguat in pediatric patients, and to assess long-term clinical outcomes such as heart failure severity and hospitalizations.

2.2 Background & Rationale To address the gap in approved, targeted therapies for pediatric heart failure (HF) due to left ventricular systolic dysfunction. While most current treatments for HF in children are extrapolated from adult approvals, rates of pediatric HF are increasing, bringing high morbidity and mortality. Vericiguat, a soluble guanylate cyclase (sGC) stimulator, has demonstrated significant efficacy in adults with worsening heart failure (VICTORIA trial). The VALOR study investigates if vericiguat can safely and effectively reduce NT-proBNP levels and improve HF outcomes in children. As of Protocol Amendment 2, the separate open-label extension arm of study MK-1242-043 (NCT06428383) is incorporated into MK-1242-036 as an optional Extension Period.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional Control Type: Placebo-controlled Blinding: Double-blind Randomization: Randomized Trial Status: Recruiting

3.2 Planned Enrollment & Duration Target N: 342 pediatric participants Number of Sites: Multicenter (Global) Estimated Study Start: 31-May-2023 Estimated Primary Completion: 15-Apr-2032

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 28 days to < 18 years inclusive. 2. Has symptomatic chronic heart failure (HF) resulting from systemic left ventricular (LV) systolic dysfunction with biventricular physiology and a morphologic systemic left ventricle. 3. Has a left ventricular ejection fraction (LVEF) $< 45\%$ assessed within 3 months before randomization, weighs ≥ 3 kg, and is currently receiving stable medical therapy for HF. 4. Is of any sex/gender. Females are eligible if not pregnant or breastfeeding, and at least one of the following: is not a participant of childbearing potential (POCBP); or is a PCBP who uses a highly effective contraceptive method, has a negative highly sensitive pregnancy test, abstains from breastfeeding during the study intervention period and for at least 30 days after, and has had their medical history, menstrual history, and recent sexual activity reviewed. 5. **Extension Period Criteria:** Was randomized, received at least 1 dose of study intervention (vericiguat or placebo), did not permanently discontinue study intervention, and completed the Week 52 visit and safety follow-up period of the Base Period.

4.2 Exclusion Criteria 1. Is clinically unstable (e.g., has symptomatic hypotension or is hypotensive for age, recent use of intravenous inotropes and/or IV vasodilators, or recent IV diuretics). 2. Has a known allergy or sensitivity to vericiguat, any of its constituents, or any other SGC stimulator. 3. Has a history of single ventricle heart disease, morphologic systemic right ventricle, hypertrophic or restrictive cardiomyopathy, active myocarditis or presumed/definitive myocarditis, or has undergone/is awaiting Class 1A heart transplantation, or is receiving continuous IV inotropes, or has an implanted ventricular assist device. 4. Has sustained or symptomatic dysrhythmia uncontrolled with drug or device therapy. 5. Has had a recent cardiovascular (CV) surgical procedure or percutaneous intervention to palliate/correct congenital CV malformations, or has unoperated or residual hemodynamically significant congenital cardiac malformations. 6. Has acute coronary syndrome, undergone recent coronary intervention, or has an indication for coronary revascularization. 7. Has symptomatic carotid stenosis or other symptomatic cerebrovascular disease. 8. Has severe pulmonary

hypertension, requires continuous home oxygen for significant pulmonary disease, or has known interstitial lung disease. 9. Has severe chronic kidney disease or a hepatic disorder (e.g., hepatic encephalopathy, hepatic laboratory abnormalities, or Child Pugh Class C). 10. Has a gastrointestinal or biliary disorder that could impair absorption, metabolism, or excretion of medications. 11. Has significant bone disease (other than osteopenia) that can alter bone formation. 12. Has concurrent or anticipated concomitant use of phosphodiesterase type 5 (PDE5) inhibitors or another sGC stimulator. 13. Has received a COVID-19 vaccination within 1 week before randomization.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Vericiguat or matching placebo administered once daily. The starting dose is based on the child's weight, with 1 to 2 dose up-titrations planned depending on the patient's blood pressure and tolerability. **Route of Administration:** Oral (Tablet or Oral Suspension at 0.2 mg/mL or 1 mg/mL). **Duration of Treatment:** 52 weeks (Base Period), followed by an optional open-label extension period.

5.2 Concomitant & Prohibited Therapy Permitted: Stable background standard of care medical therapy for heart failure. **Prohibited:** Concurrent administration of intravenous inotropes or intravenous vasodilators during the study period; therapies conflicting with sGC stimulators (e.g., PDE5 inhibitors).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, LVEF confirmation, NT-proBNP Labs
Baseline	Day 0	Randomization, First Dose, Weight/Vital Signs
Interim	Week 16	Primary Efficacy Assessment: NT-proBNP Labs, PK sampling, Safety/BP check, Dose Titration if applicable
End of Study	Week 52/58	Final Base Period Efficacy/Safety checks, ECG, bone X-rays, transition to open-label extension

(Note: Additional safety visits and blood pressure monitoring occur throughout the titration phases)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Base Period:** Change from baseline to Week 16 in log-transformed N-terminal pro-brain natriuretic peptide (NT-proBNP). * **Extension Period:** Percentage of participants with one or more adverse events (AEs). * **Extension Period:** Percentage of participants who discontinued study drug due to an AE.

7.2 Secondary & Exploratory Endpoints * **Base Period:** Change from baseline to Week 52 in log-transformed NT-proBNP. * **Base Period:** First event of cardiovascular (CV) death, heart failure hospitalization (HFH), or worsening of heart failure (HF) without hospitalization. * **Base Period:** Percentage of participants with one or more adverse events (AEs). * Pharmacokinetics (PK) profile of vericiguat in the pediatric population (e.g., AUC, Cmax). * Long-term clinical efficacy including time to HF hospitalization and changes in echocardiographic parameters.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine vital signs (blood pressure for hypotension), continuous clinical laboratory assessments, ECGs, bone X-rays, and hearing tests (for participants ≥ 6 years old). **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline. Active monitoring for HF exacerbation, cardiovascular events, and severe hypotensive episodes. **Data Monitoring Committee (DMC):** Independent External Data Monitoring Committee to regularly review unblinded safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) Set for efficacy endpoints (all randomized participants); Safety Set for all participants receiving at least one dose of study medication. **Sample Size Justification:** Target $N=342$ appropriately powered to detect a statistically significant superior reduction in log-transformed NT-proBNP at Week 16 for vericiguat compared to placebo. **Handling of Missing Data:** Mixed-Effect Model Repeated Measure (MMRM) and/or Multiple Imputation (MI) methodologies to handle missing continuous biomarker data at Week 16.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Risk-based monitoring involving regular onsite visits for PK/PD sampling verification and remote monitoring for data validation. **Audit Trail:** Robust eCRF systems with audit trails for dose-titrations, blood pressure records, and data clarification forms.

11. Study Identifiers & Governance

Primary ID: MK-1242-036 **Registry Number:** NCT05714085 **Posting ID:** 791002599937791567 **Sponsor/Lead Organization:** Merck Sharp & Dohme

LLC Study Title: VALOR **Official Regulatory Title:** A Phase 2/3 Randomized, Placebo-Controlled, Double-blind, Clinical Study to Evaluate the Efficacy, Safety, and Pharmacokinetics of Vericiguat in Pediatric Participants With Heart Failure Due to Systemic Left Ventricular Systolic Dysfunction **Trial Status:** Recruiting

12. Clinical Framework (Heart Failure Specific)

Primary Condition: Heart Failure due to Left Ventricular Systolic Dysfunction (LVSD) **Disease Area:** Heart Failure, Left Ventricular Systolic Dysfunction **Preferred UMLS Name:** Systolic Heart Failure **Semantic Type:** Disease or Syndrome **Condition Definition:** A heterogeneous condition in which the heart is unable to pump out sufficient blood to meet the metabolic need of the body. Heart failure can be caused by structural defects, functional abnormalities (VENTRICULAR DYSFUNCTION), or a sudden overload beyond its capacity. Chronic heart failure is more common than acute heart failure which results from sudden insult to cardiac function, such as MYOCARDIAL INFARCTION. **MeSH Code:** D006333 | **SNOMED ID:** 84114007 **Medical Methodology:** Background HF Therapy + sGC Stimulator (Vericiguat) **Optimization Target:** Weight-based dose titration of Vericiguat (ATC Code: C01DX22) to target exposure without causing symptomatic hypotension

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for Pediatric HF **Education Protocol (HCP):** Protocol-defined dose titration and BP monitoring training **Education Protocol (Patient):** Caregiver training on oral suspension administration and symptom monitoring **Quality Audit Mechanism:** Treatment adherence, dose-titration compliance, and NT-proBNP tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to 58 weeks (52-week base treatment + follow-up), plus optional extension **Onsite Visit Cadence:** Baseline, Week 16, Week 52, and Extension visits **Remote Monitoring Frequency:** Regular intervals for BP and safety checks during dose titration **Checklist Review Interval:** Quarterly safety and efficacy reviews

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				
Statistician	[Name]	_____	[DD-MMM-YYYY]				